CHE 201 HW 2
Due: 11.59pm on 02/02/2020

1. (20 points)

2.12 WP During the packaging process, a can of soda of mass 0.4 kg moves down a surface inclined 20° relative to the horizontal, as shown in Fig. P2.12. The can is acted upon by a constant force **R** parallel to the incline and by the force of gravity. The magnitude of the constant force **R** is 0.05 N. Ignoring friction between the can and the inclined surface, determine the can's change in kinetic energy, in J, and whether it is *increasing* or *decreasing*. If friction between the can and the inclined surface were significant, what effect would that have on the value of the change in kinetic energy? Let $g = 9.8\ \text{m/s}^2$.

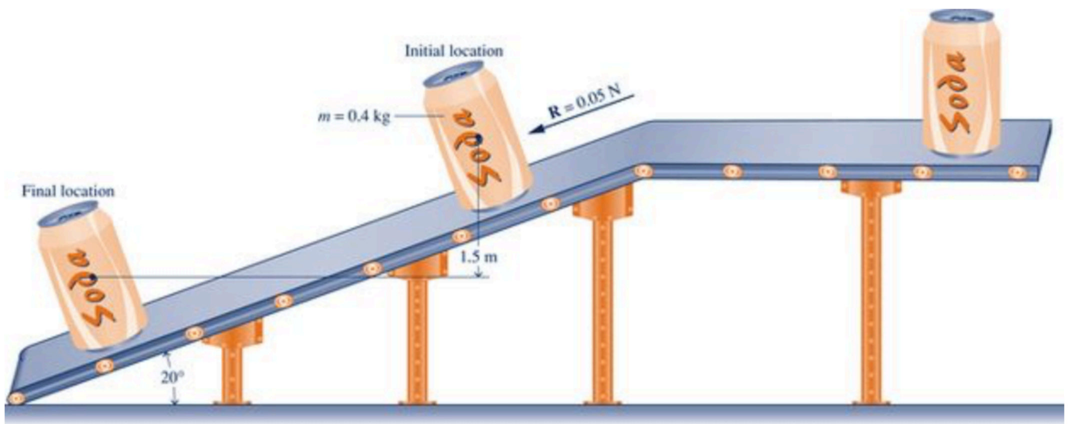

FIGURE P 2.12

2. (20 points)

2.17 WP A gas in a piston–cylinder assembly undergoes a process for which the relationship between pressure and volume is $pV^2 = constant$. The initial pressure is 1 bar, the initial volume is 0.1 m^3, and the final pressure is 9 bar. Determine (a) the final volume, in m^3, and (b) the work for the process, in kJ.

3. (20 points)

2.25 Air contained within a piston–cylinder assembly undergoes three processes in series:

Process 1–2: Compression during which the pressure–volume relationship is $pV = constant$ from $p_1 = 10$ lbf/in.^{2}, $V_1 = 4$ ft^3 to $p_2 = 50$ lbf/in.^{2}

Process 2–3: Constant volume from state 2 to state 3 where $p = 10$ lbf/in.^{2}

Process 3–1: Constant pressure expansion to the initial state.

Sketch the processes in series on p–V coordinates. Evaluate (a) the volume at state 2, in ft^3, and (b) the work for each process, in Btu.

4. (20 points)

2.31 WP Figure P2.31 shows an object whose mass is 5 lb attached to a rope wound around a pulley. The radius of the pulley is 3 in. If the mass falls at a constant velocity of 5 ft/s, determine the power transmitted to the pulley, in hp, and the rotational speed of the shaft, in revolutions per minute (RPM). The acceleration of gravity is 32.2 ft/s^2.

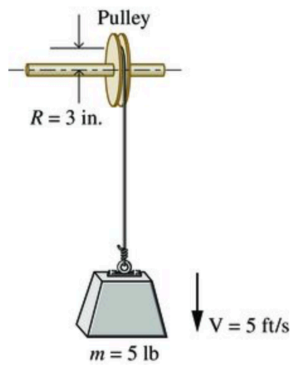

FIGURE P 2.31

5. (20 points)

Battery disposal presents significant concerns for the environment (see Energy and Environment, Sec. 2.7) Research the current federal regulations and those in your state and local area that govern the collection and management of used batteries. Write a report (maximum 2 pages) with at least three references. In your report, comment on the effectiveness of services in place to assist consumers in complying with those regulations.